

A machine learning approach for accurate crop type mapping using combined SAR and optical time series data

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Abstract

A country's food needs mainly depend upon its agriculture resources and require reliable information related to crop health, distribution, and acreage estimation to manage and monitor resources to implement a sustainable agricultural system. Different methodologies have been used to collect this information. However, the availability of earth resource satellite data with improved spatial, spectral, and temporal resolutions, such as the European Space Agency's (ESA) Copernicus program satellites Sentinel-1 (S1) and Sentinel-2 (S2), are creating more practicability to generate crop type maps. S1 and S2, both operating in a constellation of twin satellites, carry a C-band Synthetic Aperture Radar (SAR) and Multispectral Instrument (MSI). Vertical transmit and horizontal receive (VH), and vertical transmit and vertical receive (VV) channels of S1 were used to exploit the temporal backscatter of crops present in the study area. In this research, a machine learning random forest classification algorithm is used for accurate crop type mapping through the combination of SAR (S1) and optical (S2) time-series data. Random forest classifier has produced considerable improved accuracies of crop type mapping in previous studies as it uses ensemble decision trees trained on sample data which permit the vote in favor of the most popular land use/ land cover class. The key objectives of this study are to investigate the classification accuracies for different data combinations. Three plots of data are tested (i) S1 (ii) S2 (iii) S1 & S2. A combination of SAR and optical data turn out with the best overall accuracy of 97% and a kappa coefficient of 0.97. Space-borne SAR and optical data add a new aspect of crop type mapping, which increases the classification accuracy by including valuable parameters and beating the drawbacks of each other. By comparing the results, it can be concluded that combining all-weather accessible SAR and spectrally rich optical data accomplished more accurate outcomes. It would be an imperative advance for the future endeavor to estimate crop biomass and biophysical parameters.

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1. Introduction

The investigation over the provision of food to the developing world population has been debated for many years. The population grew to 6.9 billion in 2010, up from 2.5 billion to 3.7 billion in 1950 and 1970. However, per

capita food consumption has increased as world food production grows faster than the population. According to the United Nations, the total global population is expected to increase by 9.2 billion in 2050 (UNFPA, 2008). After this expansion, the farming segment will slow down compared to the past by affecting its growth rate (Alexandratos and Bruinsma, 2012). Agricultural production will need to be triple in the coming years as a shift is expected in consumption patterns and population growth (Van Beek et al., 2010; Tilman et al., 2002; Keyzer et al., 2005). Our planet

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does not yet have enough food for its nearly one million fast-growing population in the context of global agriculture and food demand. Agriculture is the most prominent land use on the planet, occupying approximately 38 percent (%) of the Earth's terrestrial surface. Globally, 12% (1.53 billion hectares) of snow-free land is cultivable, while the remaining 26% (3.38 billion hectares) of land is ice-free pasture (Foley et al., 2011).

In recent years, hunger has steadily increased as the number of people suffering from malnutrition has increased from 804 million in 2016 to around 821 million in 2017 (FAO, 2019a). One-third of food for human needs is lost or destroyed each year, which deliberately costs the environment and economy and hinders global food security goals (FAO, 2019b). Over the next 40 years, the world's hunger and growing population can be met by a 60% increase in agricultural production (OECD-FAO, 2015), and a better agrarian policy can address the food security issues more effectively. Hunger and poverty eradication was incorporated as one of the United Nations Millennium Development Goals adopted in 2000. Investment reforms on natural resource management, water resource, rural development, smallholding farming, and agriculture research could achieve effective food security policy.

Asia has the highest urban population rate of 1.5% per year. It is becoming increasingly urbanized in the current circumstances, with the expected average percentage of change in the urban population indicating that by 2030 it will exceed 55% (FAO, 2019c). Malnutrition is a threat to life in rural and urban areas. Therefore, new initiatives should be introduced to strengthen the capacity of the agricultural system in South Asia. Wheat and rice are two significant crops grown in Pakistan, India, Bangladesh, and Nepal on about 13.2 million hectares, known as South Asia (Hobbs et al., 2003; Fujisaka et al., 1994). Both these crops sustain regional food security and assume a critical role in the employment and development of the rural population. The presence of wheat, barley, and jojoba in the Indian subcontinent in 9000 BCE proves the region's ancient history (Gupta, 2004). The development of farming in the Indus basin and different zones of the Indian subcontinent resulted from favorable conditions that have been made through the increase of precipitation in this region.

The Islamic Republic of Pakistan is a sovereign country and agrarian economy in South Asia, encompassing 796,095 square kilometers (km²). Pakistan has an industrialized production economy with a well-integrated sector of agriculture. Pakistan has the world's most extensive irrigation system, and almost 80% of the country agriculture depends on it. The agriculture sector contributes to the country's Gross Domestic Product (GDP) by 21%, and 44% of the rural population is directly involved in this sector as their primary source of livelihood (Khan & Khan, 2018). At the same time, the value added in overall agriculture by a % of 23.85 and 4.66% of GDP on owing to some critical crops such as wheat, rice, maize, cotton, and sugarcane (Govt of Pakistan, 2016). The implementations of the

agriculture sector during 2016–2017 stayed up to the stamp and accomplished developments of 3.46% against the objective of 3.5%. This was conceivable by better reaping crops through more prominent accessibility of farming information sources like water, agribusiness credit, and fertilizers. Pakistan has only 8% of the United States (US) territory, and its population is expected to increase from 185 million in 2014 to 271 million by 2050, which is the current US population (Dempewolf et al., 2014). Therefore, Pakistan needs accurate estimation and forecasting of agricultural statistics, which will later yield great food security outcomes, the country's economy, and market development.

Improving the profitability of inputs while conserving resources in farm management by establishing a Decision Support System (DSS) defines precision agriculture (PA) research (Milella et al., 2019). Good use of PA came out in the 1980s (Gebbers and Adamchuk, 2010) to apply the proper treatment to the resources. PA is widely acknowledged as a pivotal contributor to the crop production as it monitors crop health, water stress, acreage and improves crop yields by using some basic predictive technologies (Liaghat and Balasundram, 2010). Remote Sensing, Geographic Information System (GIS) and Global Positioning System (GPS) are essential for precision agriculture; they divide the farmlands into management zones based on soil moisture content, crop properties, and fertilization requirements (Grisso, 2009). Remote Sensing and GIS advancements are integrated. Remote Sensing systems give satellite-based information input to GIS frameworks to analyze, interpret, and act as a DSS. Combining these systems helps formulate a mapping system that can provide better natural resource management in spatial, spectral, and temporal manner (Esri, 2008). Remote sensing technology plays a vital role in identifying crops on administrative units. Authorities need to know about crop conditions and quantity estimation of different crops in specific units (Leslie et al., 2017) for sustainable planning because of farmlands' vast size, which cannot be achieved with traditional strategies. Remote Sensing systems assist farmers in deciding planting and harvesting dates considering the weather patterns (Sacks et al., 2010) and soil types. By observing the quality and other parameters of the crop, researchers can predict the yield of farmlands before harvesting (Shanahan et al., 2001).

Improvement in digital image processing algorithms and increasing earth resource data availability made remote sensing a systematic and cost-effective mapping technology (Gómez et al., 2016). In an Agro-environmental management situation, it is advantageous to delineate plots that have just been plowed and information related to soil tillage critical to train models that need different crop parameters to estimate yield, water, and carbon fluxes (Daughtry et al., 2006). Since germinated seed arises, monitoring the emergence period can be accomplished through Normalized Difference Vegetation Index (NDVI) temporal analysis (Baghdadi and Zribi, 2016). Information derived from

remotely sensed data helps factory managers to launch harvesting campaigns. Knowledge of the dynamic use of land use/ land cover in agricultural landscapes still poses a serious challenge to the scientific community (Corgne, 2014).

In the recent course of time, machine learning algorithms (MLAs) are extensively used in remote sensing as an inductive approach for pattern recognition and their use in other similar data to predict for regression and classification (Cracknell & Reading, 2014). Random forest (RF) classifier is an ensemble classifier that usually gives more accurate results than other classifiers (Pal, 2005; Zhou et al., 2017), and researchers taking much interest as it provides robust control, speedy processing, and excellent classification results (Du et al., 2015). In contrast to traditional parametric classification algorithms like Maximum likelihood classification (MLC) or ISODATA clustering, the RF technique learns the relationship between response and training data (Breiman, 2001). It is a supervised classification method built on decision trees while conclusive results of each tree, based on bootstrapped samples of training data, are randomly selected and calculates response variables (Gislason et al., 2006). Rodriguez-Galiano et al. (2012) summarized some advantages of RF classification as follows: (1) It runs effectively on huge databases., (2) It can deal with a huge number of information factors without variable erasure, and (3) It gives appraisals of what factors are imperative in the characterization. The RF algorithms in multiple applications such as ecosystem, land use/ land cover mapping, and crop classification used by Kyere et al. (2019), Adam et al. (2014), Akar & Güngör (2015), and Millard & Richardson (2015).

The European Commission and the European Space Agency (ESA) launched a joint initiative started in 1998 known as Global Monitoring for Environmental and Security (GMES) under the Copernicus program to establish a capacity system for environmental monitoring and security applications (Delegido et al., 2011). European Space Agency's new satellites Sentinel-1, -2, -3, -5 provides support from space to monitor and control natural and artificial features by ensuring that its inhabitants are prepared and protected against all crises (Ingmann et al., 2012). Data gathered from Sentinels are analyzed by researchers in a way that provides information on past, present, and future trends (Copernicus, 2019). GEMS satellites contributing to better food management by overseeing the irrigation of farmlands and crop monitoring (Torbick et al., 2017). This study examined the combined use of Sentinel-1 and Sentinel-2 to explore the potential use of radar and optical data for crop classification. European Radar Observatory under the GEMS launched Sentinel 1A, and Sentinel 1B, a constellation that operates as an imaging radar mission and contributing to day-night and all-weather C-band imagery (ESA, 2012). Sentinel-1 acquires data in four different modes, swath widths and spatial resolutions. (Geudtner et al., 2014). It can receive data in dual and single polarization HH + HV, VV + VH & HH, VV respectively (Torres et al., 2012). Sentinel-1 data was used in agricultural mapping, damage

assessment, and soil moisture mapping (Song and Wang, 2019; Balenzano et al., 2011; Plank and Simon, 2014). The Sentinel-2 mission provides high-resolution imagery with a wide swath and constitutes twin satellite Sentinel 2A and Sentinel 2B constellation phased at an angle of 180° to provide a high temporal resolution of 5 days at the equator (ESA, 2015). Onboard, optical sensor Multispectral Instrument (MSI) covers a 290 km swath width and a broader range of wavelengths from 0.443 micrometer (μm) to 2.19 μm (Drusch et al., 2012). Ottosen et al. (2019), Vuolo et al. (2018), and Nguy-Robertson et al. (2014) explored Sentinel-2 data for crop mapping, green leaf area, chlorophyll, and crop phenology delineation.

In recent years, radar and optical data integration analysis has been published for cropping practices (Clerici et al., 2017; Forkuor et al., 2014; Joshi et al., 2016). Cloud cover is the major constraint of optical sensors as the sun's radiation cannot penetrate through clouds and leave gaps in optical imagery (Van Tricht et al., 2018). Crop phenology monitoring and mapping significantly affect if imperative data is missing in time series (Whitcraft et al., 2015). Multisensor data combination is the widely accepted strategy to overcome cloud and cloud shadow limitations (Joshi et al., 2016). Synthetic Aperture Radar (SAR) remote sensing uses an active microwave as a source of illumination for the target area (Campbell & Wynne, 2011). It is independent of solar illumination, time of the day, and weather conditions (Li et al., 2005). Microwave radiations are not affected by the clouds (Currie, 1991) because the water droplets in clouds are so much smaller than the size of wavelengths; even larger droplets of rain will not affect the longer wavelengths. A radar sensor sends a directed energy pattern to illuminate the portion of the target surface and then receive the backscatter to the instrument. Scattering of microwave energy wavelengths on a scale of few centimeters depending on the surface roughness and dielectric properties of the targeted surface (Haack, 2007). Increasing interest in the combined use of optical and SAR data in agricultural practices gives better discrimination among various crop classes and eventually leads to better accuracy and acreage estimation as it overcomes the limitations of both data types (Hütt and Waldhoff, 2018).

The research objectives of this study is to delineate and discriminate different crop types using RF machine learning classification algorithm and time-series in the study area using three scenarios: (1) Crop identification, mapping and accuracy assessment using SAR (Sentinel-1) data exclusively, (2) Crop identification, mapping and accuracy assessment using optical data (Sentinel-2) data exclusively, and (3) Crop identification, mapping and accuracy assessment using combined SAR (Sentinel-1) and optical (Sentinel-2) data.

2. Study area

Sharaqpur is a tehsil of district Sheikhpura in the province of Punjab, Pakistan. It is located 32 km towards the

northwest of Lahore city on Lahore-Jaranwala road between $31^{\circ}31'37''$ N to $31^{\circ}31'21''$ N latitude and $74^{\circ}74'6''$ E to $74^{\circ}74'2''$ E longitude, covering an area of about 373.51 km^2 (Fig. 1). It is famous for its agricultural landscape. Wheat, rice, and strawberry are the major crops produced in the area. Other than these, jawar, maize, mustard, bajra are cultivated as minor crops. Guavas, lychee, and phalsa, are the leading fruit orchards in the area. Study area experiences extreme variation in the climate. Air is exceptionally cool from mid of December to mid of March, with light to moderate rainfalls. The summer season begins in early April, and the next two months are scorching. At the end of June, the rainy monsoon embarks, and it continues during the following two-and-a-half months.

According to the 6th Population and Housing Census 2017, the total population of Sharaqpur Tehsil is 197,220 out of which 155,262 are rural residents, and 41,958 are urban residents. This research is conducted to study different crops for Rabi season (winter crops), including wheat, fodder, and strawberry as a major crops.

The phenology of the wheat cycle was identified by several different date datasets that distinguish it better than other standing crops of sugarcane and fodder. In the study area, wheat fields were empty in October and early November, and during the month of February, rapidly growing phase dominated the wheat crop. Spectral signatures of other standings, such as fodder mixed with wheat, create

an ambiguity in discrimination. However, wheat is at its peak vegetative phase shows a different structure and vigourousness compared to fodder. The phenology of other crops helps to understand the cropping patterns of the study area. The cropping calendar for different crops present in the area is shown in Fig. 2.

3. Data and image pre-processing

3.1. Sentinel-2

Sentinel-2 (S2) provides critical information for agriculture monitoring. Ten cloud-free images of the T43TRD grid were retained from the Sentinel-2A/2B onboard MSI (MultiSpectral Instrument) sensor according to the crop stages. The baseline information was collected through surveys and interviews of local farmers. This study is conducted for the winter crop season. Wheat is the dominating crop in the region that starts from November to April and covers six months, while other crops, including fodder, strawberry, and some vegetables, are also present during the same time (Table 1). Therefore, data was acquired according to the crop sowing, growing, and harvesting stages, while the redundancy of the data was critically considered as there was no rapid change in the agricultural landscape for classification purposes. European Space Agency (ESA) operated Copernicus Open

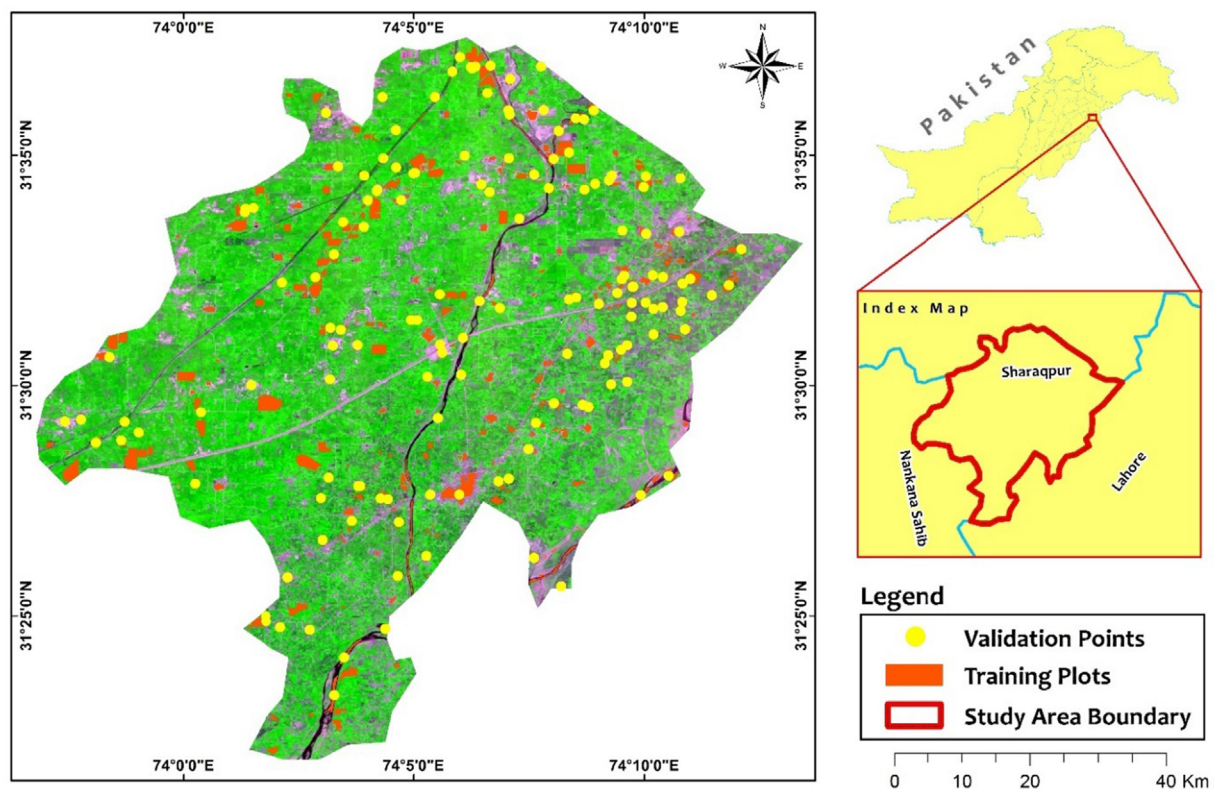


Fig. 1. Study site based on Sentinel-2 multispectral false-color composites (Red, NIR, Blue). The map also shows the location and distribution of validation points and training plots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

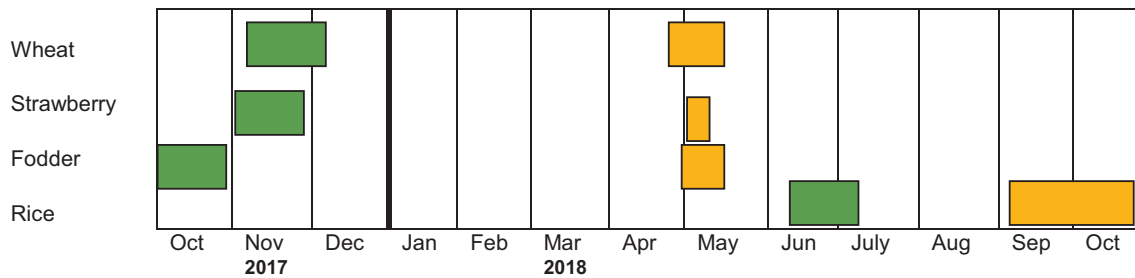


Fig. 2. Cropping calendar surveyed for each of the crops presents in the study site. Green bars represent the start of the cultivation period, and yellow bars represent the harvesting period. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Data acquisition dates of the Sentinel-2 images.

No	DOY (2017–18)	Δ Days	Dates
1	328	–	24 November 2017
2	353	26	19 December 2017
3	13	26	13 January 2018
4	21	09	21 January 2018
5	41	21	10 February 2018
6	56	16	25 February 2018
7	68	13	09 March 2018
8	81	14	22 March 2018
9	98	18	08 April 2018
10	116	19	26 April 2018

Access Hub to download the time series imagery. Level 1C (L1C) Universal Transverse Mercator (UTM)/ World Geodetic System 1984 (WGS84) orthorectified products were selected, which provide Top of the Atmosphere (TOA) per pixel radiometric measurements. Geometric corrections re-project the data into the desired projection system, which preserves the size, shape, area and incorporates the Earth curvature effect. Radiometric correction deals with column line dropouts, line starting problems, and atmospheric effects (Lillesand et al., 2015). The pre-processing involved the atmospheric correction of S2 imagery using the Sentinel Application Platform (SNAP) toolbox plugin SEN2COR (ESA, 2019c; Li et al., 2018). It converts the Level – 1C TOA to the Surface Reflectance (SR) (Level – 2A). All bands of S2 were re-sampled to 10m resolution using the nearest neighbor method to ensure the transfer of each value to the corresponding pixel accurately. Finally, the re-sampled bands were masked over the study area boundaries and combined to generate a single multi-temporal stack.

3.2. Sentinel-1

Sentinel-1A (S1) datasets were also downloaded from ESA Copernicus Open Access Hub. A cross-pol channel is when you transmit horizontal polarization but receive vertical echoes in the back or transmit vertical polarization and receive horizontal echoes coming back. The cross-pol channel is valuable when looking at the land cover mapping and crop classification (Forkuor et al., 2014) because

volume scattering from vegetation depolarize the signal considerably and receive a lot of cross-pol coming back from the scene when you are observing vegetation (De Wit and Clevers, 2004). Ground Range Detector (GRD) product's interferometer wide (IW), dual-polarization VV + VH were selected for this study (Table 2). IW, the default and main primary operational mode of S1 over the land area (ESA, 2013), and dual-polarization VV + VH interact more strongly with agricultural fields. Dew appears on crops in the morning and causes them to absorb C-band microwave radiation. Therefore, it is preferable to use an ascending orbit to prevent absorption (Gillespie et al., 1990). We have to choose a product that gives information in square pixels equal to S2 data pixels. GRD products were used by Veloso et al. (2017) and Sonobe et al. (2017) for agriculture applications and have an almost square resolution (20×22).

Although it has the disadvantage of not containing phase information, it does not make any difference as classification purposes can merely be served by backscattering except in the case of polarimetric decompositions. One of the critical advantages of using this product is to combine effortlessly with S2 by pixel spacing of 10×10 m. Pre-processing of S1 images was performed in SNAP, and the chain of pre-processing steps involves the radiometric correction, speckle filtering, and terrain correction. Intensity bands (VV and VH) were converted to normalized radar cross-section coefficient or sigma naught or original reflectivity of ground (σ_0) by applying radiometric calibration, which transformed Digital Numbers (DNs) values to radar

Table 2
Data acquisition dates of Sentinel-1 images.

No	DOY (2017–18)	Δ Days	Dates
1	329	–	25 November 2017
2	353	25	19 December 2017
3	12	25	12 January 2018
4	25	20	25 January 2018
5	44	13	13 February 2018
6	56	16	25 February 2018
7	68	13	09 March 2018
8	85	18	26 March 2018
9	97	10	07 April 2018
10	116	23	26 April 2018

backscatter (Filipponi, 2019). Transmit power, system loss, and antenna gain introduce some errors and noise in SAR captured data. SAR is a coherent imaging system enduring a common problem called speckle and is produced by the large and rough surfaces whose scale is not matched with the corresponding wavelength. Lee filter was used for this research as previous studies shows that it gives better performance to reduce speckle noise compared to other filters (Lee, 1981; Lee et al., 2008; Meenakshi and Punitham, 2011). Lee sigma filter parameters for speckle filtering are shown in Table 3.

Radar shadow is introduced when some part of the signal returns and some part was obscured by the topography, which means there is no signal coming back from the region behind the object. Another case that can happen was that the features on the ground were not in their correct geographic locations in an image. This leads to foreshortening. The layover effect incorporates a shift in the top and bottom of some objects. Range Doppler Terrain Correction rectifies these problems using Shuttle Radar Topography Digital Elevation Model (SRTM-DEM) with spatial resolution of 3 arc seconds. Geo-coding of original data was done at UTM zone 43/WGS84 and bilinear interpolation method was used for re-sampling (Tavares et al., 2019; Haas and Ban, 2017). Table 3 shows the terrain corrected parameters adopted for this study. We need a correspondence between both datasets. It is impossible to acquire all images of both sensors on the same date because cloud cover and different temporal resolution make it impossible. It was knowing the following reasons there should be a margin of ± 7 days—corresponding dates of both sensors selected to conduct this research as shown in Table 4.

3.3. Training and validation data

A field survey of the study area Tehsil Sharaqpur was carried out during the growing season of Rabi crops on two continuous days i.e., 25 February 2018 to 26 February 2018. Time and dates were picked according to the crop situation and management practices in the study site. The

Table 3
SAR speckle filter and terrain correction parameters.

Lee filter parameters	
Parameter	Value
Filter	Lee
Filter Size X	3
Filter Size Y	3
Number of Looks	1
Terrain correction parameters	
Parameter	Value
Digital Elevation Model	SRTM 3sec
Image Resampling Method	Bilinear Interpolation
Pixel Spacing	10 m
Map Projection	UTM zone 43/WGS 1984

Table 4
Corresponding dates of Sentinel-1 and Sentinel-2.

No	Satellites	$\pm \Delta$ Days	Date (2017–2018)
1	S1	1	25 November 2017
	S2		24 November 2017
2	S1	0	19 December 2017
	S2		19 December 2017
3	S1	−1	12 January 2018
	S2		13 January 2018
4	S1	4	25 January 2018
	S2		21 January 2018
5	S1	3	13 February 2018
	S2		10 February 2018
6	S1	0	25 February 2018
	S2		25 February 2018
7	S1	0	09 March 2018
	S2		09 March 2018
8	S1	4	26 March 2018
	S2		22 March 2018
9	S1	−1	07 April 2018
	S2		08 April 2018
10	S1	0	26 April 2018
	S1		26 April 2018

study area was divided into four different blocks, and from each block, using an unaligned systematic random sampling approach, the points for training and validation were marked. A GPS device Etrex 20x was used to mark the location of various crops, and 300 points of different crops including wheat, fodder, strawberry, guava orchards, and vegetables were collected. Ancillary data of crop calendar and crop stages was obtained from baseline surveys and through local farmer's interviews. Fig. 3 shows the photographs of different crop sites that were taken along with their locations. Training data directly influences classification accuracy; it builds a relationship between pixel values of each spectral band and response classes. Crop location surveyed points were almost cut into half, first portion to train classifier and the remaining to validate the classification results. The selected training dataset was used to extract the spectral signatures of various crops from S2 and overlaying analysis of multi-temporal imagery generated 244 new points, precisely and accurately. A total of 397 training polygons were digitized, and each polygon contains information of a regional administrative unit, crop name, crop code, time and date, elevation, and point id. The remaining half of the field surveyed points were used as validation data as shown below in Table 5.

4. Methodological approach

In this section, the method adopted throughout the research has been discussed. The selected method investigates that SAR when combined with optical data, can increase the classification accuracy or not. To get the results, a polygon-based RF classification approach was performed, and the accuracy of the results was assessed later. Table 6 shows experimental designs for data combinations in crop classification.



Fig. 3. Crops images captured during the field visit of the study area.

Table 5

Distribution of validation and training fields used in crop mapping. Validation data is not available for vegetables and other classes.

Crop	Validation	Training (Field Survey)	Training (New Generated)
Wheat	28	28	79
Fodder	28	28	69
Strawberry	10	15	02
Forest	10	08	07
Orchards	22	22	43
Buildup	19	14	34
Barren Land	17	17	0
Water	13	07	10
Vegetables	N/A	8	0
Other	N/A	6	0

Table 6

Experimental designs for crop mapping.

Experiment (α)	Sentinel-1 (VV & VH) amplitude
Experiment (β)	Sentinel-2 (BOA) reflectance
Experiment (γ)	Sentinel-1 amplitude and Sentinel-2 (BOA) reflectance

In an experiment (α), ten S1 images were acquired from November to April, and a maximum of two images from each month was used for classification. Experiment (β) considered using all available time-series S2 images in the same duration as S1, and experiment (γ) assessed the accuracy for crop classification when all available S1 and S2 images combined were used. S1 dataset was comprised of 20 dual polarimetric bands and S2 100 bands from 2 to 8A and 11 to 12. The stacked dataset for experiment (γ) contains total bands of experiment (α) and experiment (β). The SAR data was affected by the very low and high contrast of reflectance caused by the speckle effect, so the pixel-based classification approach performs well (Cord et al., 2010), leading to a decrease in the classifier's performance. A polygon-based approach was tested for this study as used in previous studies (Tso and Mather, 1999; Aplin and Atkinson, 2004) concluded to perform better than the per-pixel approach. The polygon-based approach uses the information at the field level and pixel value fallen in polygons was extracted and saved in the table. All the pixels in the polygon were classified in their respective class, and their relationship to other classified was established via the RF classifier. The pixels were then segregated into their predicted classes. Fig. 4 shows the pre-processing and adopted methodological flowchart of this research.

4.1. Random forest classification

The RF classification algorithm was run for the classification scheme. A comprehensive R Archive Network (CRAN) RF package version 4.6–14 was used to perform the classification. Classification code was written in R Studio using the following libraries (i) sp (ii) raster (iii) rgdal (iii) map tools (iv) RF. The process starts from loading all the datasets to R and then generating a table of training data and each dataset mentioned in the classification scheme. The table contains a column of each band pixel value to the corresponding training data polygons class code. Three hundred ninety-seven polygons were used to extract the band values.

In this process, image band data generates the RF decision trees and training data to do the split. By default, 500 trees were created. The RF classification package was used to train the model that classifies each pixel. Both Boosting and Bagging (collection of trees based on structured classifiers) are well-known parameters of RF classifiers (Rodriguez-Galiano et al., 2012). Final results were either concluded by voting of each decision tree, and previous studies proved that RF classifier has high prediction accuracy (Zhou et al., 2017). To start the algorithm, the user will provide two parameters, N number of training samples

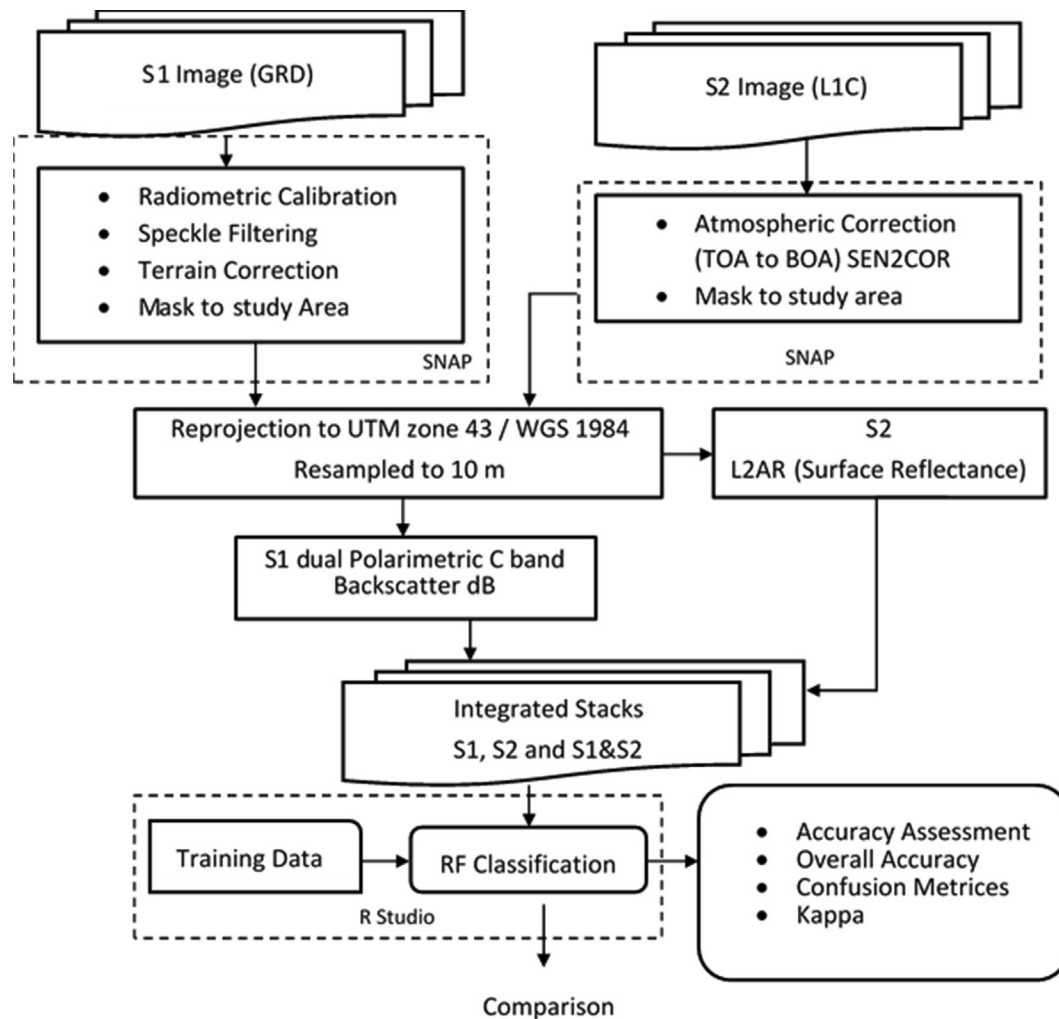


Fig. 4. Flow chart of methodological approach.

and M number of variables. The bagging approach selected N number of samples randomly to form a tree and M number of variables chosen randomly at each node used to split to create sub-nodes (Ok et al., 2012). The number of trees that a user wants to create is the first variable, and it can be selected as large as desired, but error can be controlled up to 500 trees (Cutler et al., 2007). The GINI index is used to find the best variable for splitting the node (Watts and Lawrence, 2008). GINI index notes the class homogeneity and heterogeneity and is directly proportional to it; as the GINI index increases, class homogeneity increases and vice versa. Zero GINI index shows the completion of splitting, and if a child node of the GINI index is smaller than the parent, it shows the split is successful (Watts et al., 2011).

4.2. Accuracy assessment

The measure of agreement between a standard, assumed to be correct, and a classified image of unknown quality defines the accuracy of classification (Campbell and Wynne 2011). To assess the accuracy, it is necessary to

compare the classified data with the reference data of the same sites (Jensen and Im 2007). Accuracy is required for maps and reports derived from satellite data and their operational utility for ground management. Several factors cause classification errors, such as misassignment of information classes to spectral classes, opted algorithm, image pre-processing, and incorrect selection of training sites. It is supposed that all dissimilarities in classification maps and selected training sites induce errors in the classification process. The confusion matrix or error matrix is a commonly used technique for classification accuracy assessment. The matrix shows a comparison between the correlation of sampling data and classification results. Overall accuracy is one of the standard measures of classification accuracy, which gives the proportion of samples that have been correctly classified, computing by the ratio of the sum of the diagonal entries to the total number of pixels under investigation (Campbell and Wynne, 2011). How accurate the map is from the map maker perspective is called producer accuracy, while the user accuracy indicates how accurate the map is from the user point of view. The kappa coefficient of agreement (KAPPA) is another

parameter used to analyze error matrices (Lunetta and Lyon, 2004). The KAPPA is frequently used to summarize the outputs of an accuracy assessment and is utilized to evaluate the land use/ land cover classifications acquired by remote sensing. Accuracy assessment was computed on a total number of 147 field points spread evenly on the study site.

5. Results

5.1. Results of crop type mapping

The methodology discussed was implemented on SAR and optical datasets using the optimum classifier parameters and several classified maps of the entire study area were produced from different combinations of datasets. The classification color scheme is adopted from the National Land Cover Dataset Classification (2016), various colors were selected to classify map legends. Polygon-based RF classification technique was used to classify the imagery contained in experiments (α , β , and γ) for crop type mapping. The experiment (α) dual polarimetric (VV and VH) time-series data, from sowing to harvesting period, gave different backscatter values that help understanding the relation between crop phenology and SAR backscattering values. The temporal backscatter signature of the VH channel for wheat, fodder, and strawberry is shown in Fig. 5. Wheat crop showed a low reflectivity in the months of sowing and harvesting and a maximum of 22–27 decibel (dB) in the peak vegetation stage. The experiment (α) gave an overall accuracy of 87.07% and KAPPA value of 0.84, whereas experiment (β) bottom of the atmosphere (BOA) showed an increase in accuracy with an overall accuracy of 95.34% and KAPPA value of 0.95 (Table 7). The last experiment (γ), which was the fusion of both types of SAR and optical data, demonstrated the most effective results with an overall of 97.96% and KAPPA value of 0.97 resulted in an increment of 8.31% and 2.5% in overall accuracy compared to experiment (α) and experiment (β) respectively (Table 7).

Table 7

Results of crop type mapping based on RF classifier for the three tested experiments.

Experiments	Data	OA%	Ka
(α)	S1	87.07	0.84
(β)	S2	95.34	0.91
(γ)	S1 + S2	97.96	0.95

Fig. 6 shows all the maps of the above three experiments, and it can be observed that wheat is the most abundant crop type in the entire study area. Fodder is another abundant crop attached to the tiny fields and scattered throughout the study area. Other important crops such as strawberries, fruit orchards, and vegetables were also present in the southeastern part of the study area. Field surveys showed that most of the wheat crops found in the northwestern region lie on the site that is off from the major roads and some of the agricultural land is also affected by salinity and water-logging. Therefore, farmers are reluctant to cultivate fruits and vegetables in the northwestern part of the study area and prefer the more fertile southeastern part for the crop cultivation that can easily be accessed through the major roads to deliver them to significant city markets. All three maps, produced from the study site's classification, clearly shows the exact spatiotemporal distribution of crops in-line with the interviews of the local farmers during field visits and baseline surveys.

5.2. Accuracy assessment

Classification accuracy is performed for each experiment as shown in Table 6, which summarizes the results of the crop type mapping.

From November to April, the experiment (α) was performed on twenty VV and VH polarizations bands and achieved an overall accuracy of 87.07%, lowest in all three experiments. There is considerable confusion between different classes, especially some validation pixels of fodder classified in wheat, reducing the fodder producer's accuracy

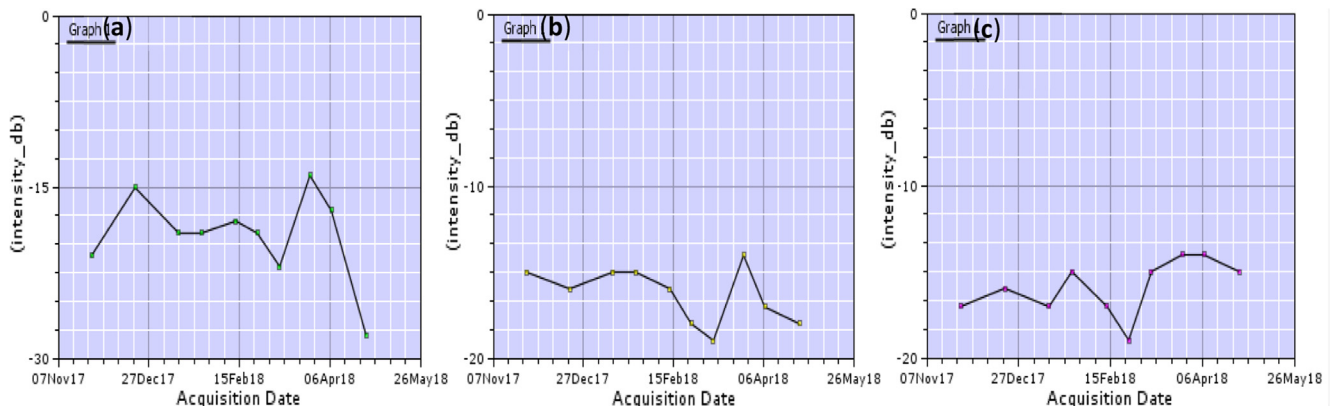


Fig. 5. Graph plots for temporal backscatter of VH polarization of (a) Wheat (b) Fodder (c) Strawberry.

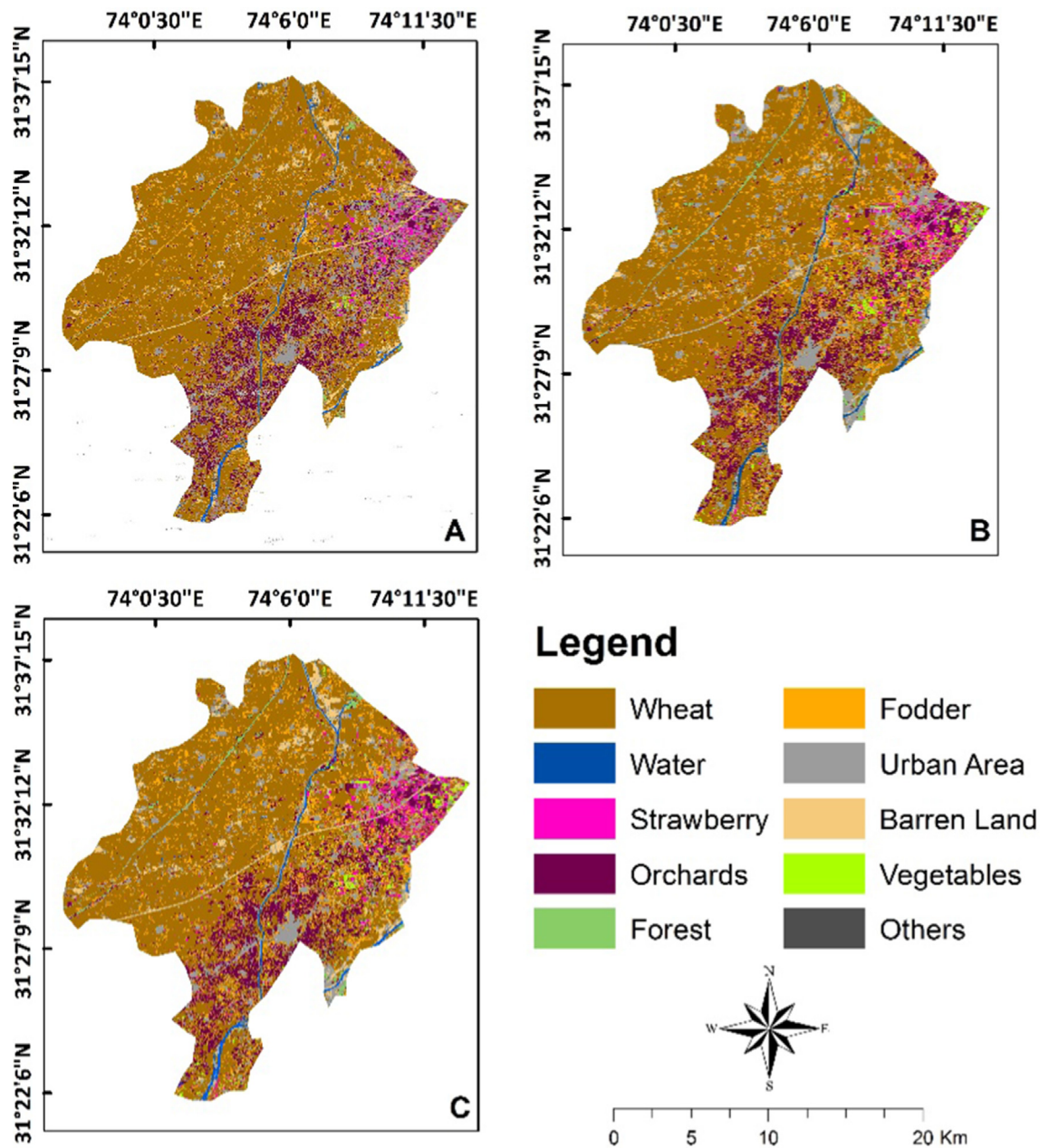


Fig. 6. Crop classification maps for Sharaqpur region in Punjab, Pakistan, produced using the RF classifier. Where (A) is for SAR (S1) only, (B) is for optical (S2) only, and (C) is for S1 and S2 combined.

by 67% and wheat user's accuracy by 82% (Table 8). The experiment (β) conducted on one hundred BOA bands resulted in an overall accuracy of 95.34%, which was much higher than the previous experiment and showed less confusion among different classes. A small number of pixels were classified in wheat and other classes, which raised the correct points to 25 out of 28 points. The producer's accuracy of fodder and user's accuracy of wheat consequently increased to 89% and 100%, respectively (Table 8). Other classes, such as urban areas and barren lands, showed more mixing into one another. Compared to experiment (α), the producer's accuracy of urban land decreases to 5%, and the user's accuracy of barren land was 6% in

response to the experiment (α). Tables 9–11 shows confusion matrices for all three experiments.

In the experiment (γ), the use of all available SAR and optical time series (November, December, January, February, March, and April) resulted in an overall accuracy of 97.96%, the highest among all three experiments (Table 8). Comparing to the experiment (α) and experiment (β), the fodder class in the experiment (γ) expressed a total of 26 corrected points out of 28, and the producer's accuracy of fodder increased to 25% and 3%, respectively. Confusion matrix produced from accuracy assessment of combined datasets showed a very little uncertainty observed among urban area and barren land classes, e.g., only one point

Table 8

Producer's accuracy (PA) and user's accuracy (UA) for all land cover classes for the Sentinel-1, Sentinel-2 and combined using RF classification.

Classes		Exp (α)	Exp (β)	Exp (γ)
Wheat	PA	100%	100%	100%
	UA	82%	90%	93%
Water	PA	100%	100%	100%
	UA	86%	100%	100%
Strawberry	PA	90%	100%	100%
	UA	100%	100%	100%
Orchards	PA	86%	100%	100%
	UA	95%	100%	100%
Forest	PA	90%	90%	100%
	UA	90%	100%	100%
Fodder	PA	67%	89%	92%
	UA	100%	100%	100%
Urban Area	PA	94%	89%	94%
	UA	69%	94%	100%
Barren Land	PA	76%	100%	100%
	UA	100%	94%	100%
Overall Accuracy		87.07%	95.34%	97.96%

was misclassified (instead of urban area it fell in the barren land), which increased its producer's accuracy. Likewise, with fodder and wheat, less misclassification increased their accuracy compared to the previous two experiments.

Table 9

Confusion matrix for the experiment (α) final map accuracy.

Classified	Reference Data								Total
	Wheat	Water	Strawberry	Orchards	Forest	Fodder	Urban	Barren	
Wheat	28	0	0	0	0	5	1	1	34
Water	0	13	0	0	0	0	0	2	15
Strawberry	0	0	9	0	0	0	0	0	9
Orchards	0	0	0	19	0	1	0	0	20
Forest	0	0	0	0	9	1	0	0	10
Fodder	0	0	0	0	0	19	0	0	19
Urban	0	0	1	3	1	2	18	1	26
Barren	0	0	0	0	0	0	0	13	13
Total	28	13	10	22	10	28	19	17	147

Table 10

Confusion matrix for the experiment (β) final map accuracy.

Classified	Reference Data								Total
	Wheat	Water	Strawberry	Orchards	Forest	Fodder	Urban	Barren	
Wheat	28	0	0	0	0	3	0	0	31
Water	0	13	0	0	0	0	0	0	13
Strawberry	0	0	10	0	0	0	0	0	10
Orchards	0	0	0	22	0	0	0	0	22
Forest	0	0	0	0	9	0	0	0	9
Fodder	0	0	0	0	0	25	0	0	25
Urban	0	0	0	0	1	0	17	0	18
Barren	0	0	0	0	0	0	2	17	18
Total	28	13	10	22	10	28	19	17	147

6. Discussion

This study shows the combined use of freely accessible multisensor and time-series data (S1, S2), and results obtained from the experiments (β) and (γ) show the inclusion of SAR data that increased the classification accuracy. Hütt et al. (2016) and Azar et al. (2016) reported the usefulness of combining the optical and microwave remote sensing data for agriculture and land use/ land cover mapping. It is observed that the SAR system has peculiar characteristics than the optical system as it is coherent in nature, faces interference from scatters that present on the surface of the Earth, and is affected by the salt and pepper effect. The crops in the study area have very limited boundaries except for the dominant wheat crop; even though it has small fields, its extensive area has subdued it. Some crop fields, such as fodder and vegetables, comprised of only a few square meters, and hence, using low-resolution data these are very difficult to classify accurately. In experiment (α), wheat, water, and strawberry validation data show an excellent agreement with the classified map and overall accuracy of 87.07% and a KAPPA value of 0.84 (Table 7). Some fodder pixels were mixed with the wheat class and the final classified map cannot show distinct field boundaries and tends to random roughness (Fig. 7). The class distinction for S2 has been improved in different classes, especially

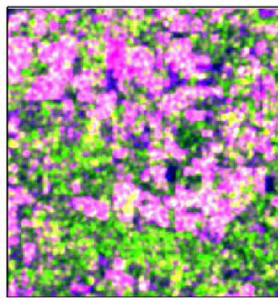
Table 11
Confusion matrix for the experiment (γ) final map accuracy.

Classified	Reference Data								
	Wheat	Water	Strawberry	Orchards	Forest	Fodder	Urban	Barren	Total
Wheat	28	0	0	0	0	2	0	0	30
Water	0	13	0	0	0	0	0	0	13
Strawberry	0	0	10	0	0	0	0	0	10
Orchards	0	0	0	22	0	0	0	0	22
Forest	0	0	0	0	10	0	0	0	10
Fodder	0	0	0	0	0	26	0	0	26
Urban	0	0	0	0	0	0	18	0	18
Barren	0	0	0	0	0	0	1	17	17
Total	28	13	10	22	10	28	19	17	147

the confusion between our main classes like fodder and wheat was reduced (Table 10). Red edge bands of S2, specially designed for crop monitoring (Delegido et al., 2011), improved spatial resolution, and 100 bands of time series were beneficial to RF algorithms in predicting the accurate number of classes. In experiment (β), the classification results, as reported in Table 7, showed an overall accuracy of 95.34% and a substantial increase (0.84 vs. 0.95) in KAPPA statistics. The accuracies enhanced with more spectral bands and high spatial resolution, and Table 8 shows the increase in producer's and user's accuracies of all classes. Fig. 7 shows the inter-field heterogeneity, indicating the field boundaries were more delineated than the experiment (α) results. Vegetation structures are more sensitive to the SAR sensor, while the onboard MSI sensor is more sensitive to the chlorophyll content (Mercier et al., 2019).

Some crops in our study site, such as wheat and fodder, have similar structures. Thus, the low accuracy of these crops was reported when observed using S1. It is also proven that L-band is more suitable for crop monitoring than C-band because it cannot penetrate through the canopy of the plants (Patel et al., 2006). However, the classification based on polarimetric indices texture and coherence of S1 data could improve the accuracy (Inglada et al., 2016; Zhou et al., 2018). The experiment (γ) results also show an increase in the producer's and the user's accuracies, and the confusion matrix shows a higher number of correct points than the previous experiments (α) and (β). The combined SAR and optical imagery dataset have significantly improved the classification accuracy and gave the highest agreement between the ground and classified data. The gains of a combination of SAR and optical data for the study area are +9.8% and +2.7% comparing to the SAR

S1 (R: VV; G: VH; B: VV/VH)



S2 (R: B4; G: B8; B: B2)

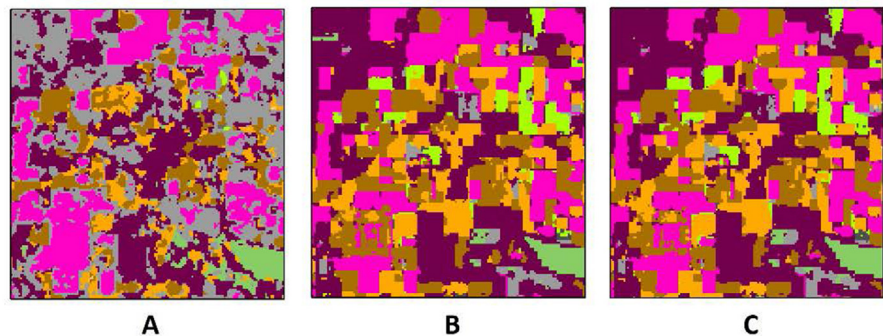
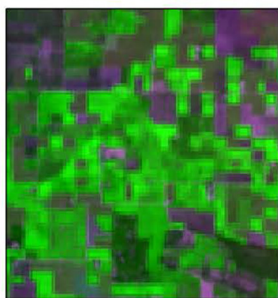


Fig. 7. Inter-field heterogeneity maps show the random roughness and extraction of field boundaries (A) SAR only (B) Sentinel-2 only (C) Combination of SAR and Sentinel-2. Classified results from A carried more random roughness and poor field boundaries delineation between different crops where B and C show less roughness and better field boundaries extraction.

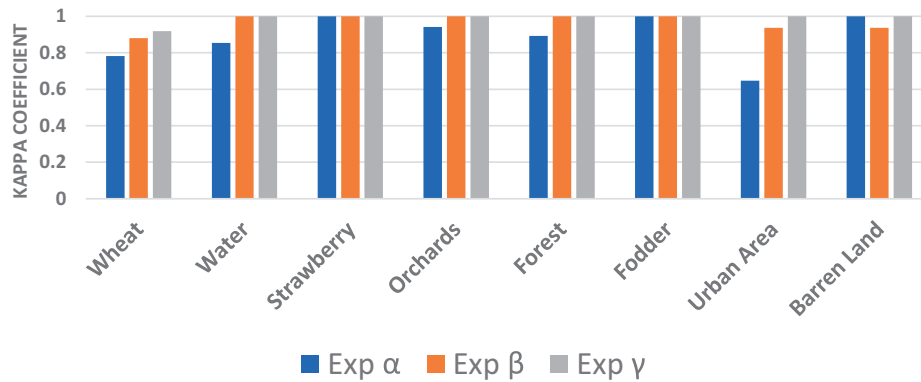


Fig. 8. Comparison of the KAPPA statistics achieved for the various crops in the study area using all three experiments.

and optical data alone, respectively. The experiment (γ) results gave an overall accuracy of 97.96% and a KAPPA value of 0.97 (Table 7), while Fig. 8 is a comparison plot for different experiments according to the different classes. Combining the properties of both types of sensors, such as dielectric and roughness properties (Joshi et al., 2016) of SAR and high spatial and spectral properties of S2, increases the accuracy. Liu et al. (2018) tested the integration of S1 and S2 for Belém in Eastern Amazon, Brazil, using the RF classification algorithm. They reported a sufficient increase of 1.4% and 1.54%, respectively, in overall accuracy when combined with the optical and SAR imagery. Sun et al. (2019) noted an increment of 4% in overall accuracy for crop type mapping while using the same datasets and classifier, as used in this study. The S1 and S2 alone showed a decrease of 17% and 1%, respectively compared to the combined optical and SAR data results. Other studies show a gain of 5% and 8% when the multi-source data is merged (Blaes et al., 2005; Shang et al., 2009). Zhang & Xu (2018) also experimented with the RF classification for SAR and optical data classification. They used SAR images of TXS/ENVISAT ASAR while optical data of SPOT-5 and Landsat Thematic Mapper (TM) sensor. The authors concluded that the best results were obtained for the RF classification, while the SAR and optical data combination increased the overall accuracy by 10%. Whyte et al. (2018) tested the RF and SVM classifiers in the synergy of S1 and S2 and found the best results when using the RF classifier and reported an overall accuracy of 83.3% and KAPPA coefficient of 0.72. Mercier et al. (2019) discussed the use of SAR and S2 alone and concluded that sometimes S2 data could be alone sufficient for classification, but S1 backscatter data also provide some important additional information. The authors reported the KAPPA coefficient of 0.88 for combined S1 and S2 and 0.86 for S2 data alone.

Optical datasets are rich in spatial and spectral information, and SAR has advantages such as high sensitivity to the dielectric properties surface roughness and the ability to achieve all-weather and day and night imagery (Mahyoub et al., 2019). The study found that the blending of inherited complimentary characteristics of both sensors

was beneficial and resulted in a better set of information and improved accuracy.

7. Conclusion

This study explored the potential of multi-temporal SAR and optical datasets in crop classification for Tehsil Sharaqpur in Pakistan. We applied RF classification algorithms in the time series of both sensors to evaluate the classification accuracy for accurate crop type mapping between 2017–2018. The microwave remote sensing has an advantage in cloudy conditions, while optical remote sensing is always affected by cloud cover. Research findings demonstrated the joint use of S1 and S2 and led to the following conclusions:

- In a complex agriculture region, the use of S1 data produced satisfactory results. However, the validation data showed a weaker agreement (Overall accuracy = 87.07% and KAPPA coefficient = 0.84) with the classified map than the optical data alone.
- The single-use of S2 reported higher accuracy than S1 data. Red edge bands of S2 were more sensitive to the vegetation than C-band of SAR data. Thus, the RF classification of the optical time-series data significantly increased the accuracy by 9.8%, resulting in an overall accuracy of 95.34% and a KAPPA coefficient of 0.95.
- The synergy of SAR (S1) and optical data (S2) produced optimal crop mapping results. The classification results of the combined dataset showed a 2.7% gain in the overall accuracy. The assessment of joint use of S1 and S2 dominated the classification results with an overall accuracy of 97.96% and a KAPPA coefficient of 0.97

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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